Software Requirements Specification

for

CareConnect

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

This document describes the software requirements for CareConnect: Smart Elder Assistance and Trusted Service Provider Management Platform. CareConnect is a web-based platform designed to help elderly people who need assistance in their daily lives. Many adult children are busy with their jobs, studies or other responsibilities and may not always be able to stay close to their elderly parents. This can make it difficult for them to ensure that their parents receive proper support and care. The main purpose of CareConnect is to connect elderly people and their family members with trusted and verified service providers. The platform allows family members to stay informed about service activities and monitor the support received by their elderly parents.The system includes features such as service requests, provider verification, family monitoring, SOS emergency alerts, video calling, payment tracking, ratings, complaints and platform administration. CareConnect aims to make elderly care easier, safer and more reliable. This SRS covers the initial university project prototype. Features such as a mobile application, live payment gateway integration, medical diagnosis and ambulance services are outside the scope of the current version and may be added in the future.

1.2 Document Conventions

This document follows some of the conventions that are described below.

1.2.1 The word “shall” is used to describe a required function of the system.
1.2.2 Each requirement has a unique ID, such as FR-01 or REQ-1.1, so that it can be easily identified and tracked.
1.2.3 Requirements may have a priority level: High, Medium or Low, based on their importance.
1.2.4 Bold text is used for important headings, requirement IDs and table headings.
1.2.5 The term “TBD” (To Be Determined) is used when some information has not been finalized yet and will be decided later.

1.3 Intended Audience and Reading Suggestions

This document is written for different people who are involved with the CareConnect project.

The intended readers are:

1.3.1 Course instructors and evaluators: They can use this document to understand and evaluate the project requirements.
1.3.2 Student project team: The project team will use this document as a guide during the design, development and testing of the system.
1.3.3 Testers: They can use the system requirements to prepare and perform test cases.
1.3.4 UI/UX designers: They can use the requirements to design the user interface of the system.

1.3.5 Future developers or students**:** They can use this document if they want to continue or improve the project in the future.

Readers who want to understand the project generally can start with the Overall Description section. Readers who need detailed information about the system should read the Functional Requirements and Non-Functional Requirements sections. It is recommended that all readers read the Introduction and Overall Description before moving to other sections.

1.4 Product Scope

CareConnect is a web-based platform that helps elderly people and their family members find trusted and verified service providers for daily assistance and care-related needs. Many people are busy with their jobs, studies or other responsibilities and may not always be able to take care of their elderly parents personally. CareConnect aims to help them find reliable service providers and stay informed about the services received by their elderly family members. The main features of the system include service provider verification, service requests, provider matching, family monitoring, SOS emergency alerts, video calling, payment tracking, ratings, complaints, and platform administration. CareConnect aims to make elderly care easier, safer and more organized. It helps family members stay connected with elderly people even when they cannot always be physically present. CareConnect is not a hospital, medical service or ambulance service. The system does not replace doctors or professional medical care. Its main purpose is to connect users with trusted service providers and improve communication, monitoring and safety. This project is being developed as a university software engineering project. The current version focuses on the main features of the system and may be improved with additional features in the future.

1.4 References

1.4.1 IEEE Std 29148-2018, *Systems and Software Engineering — Requirements Engineering*.
1.4.2 IEEE Std 830-1998, *Recommended Practice for Software Requirements Specifications*.
1.4.3 Laravel Framework Documentation.
1.4.4 MySQL 8.0 Reference Manual.
1.4.5 Bootstrap 5 Documentation.
1.4.6 Web Content Accessibility Guidelines (WCAG) 2.1, W3C.
1.4.7 OWASP Top Ten Web Application Security Risks

2. Overall Description

2.1 Product Perspective

CareConnect is a new web-based application developed as a university project. It is designed to help elderly people and their family members find trusted and verified service providers.The system makes it easier for families to manage elderly care and stay informed when they cannot

always be physically present. CareConnect includes service management, provider verification, family monitoring, safety alerts and communication features.

2.2 Product Functions

The main functions of CareConnect are:

2.2.1 User registration and login
2.2.2 Role-based access for Elderly Users, Family Members, Service Providers, and Administrators
2.2.3 Service Provider registration and verification
2.2.4 Creating and managing service requests
2.2.5 Matching users with suitable Service Providers
2.2.6 Family monitoring of elderly users
2.2.7 SOS emergency alerts
2.2.8 Video calling between Elderly Users and Family Members
2.2.9 Payment tracking
2.2.10 Ratings, reviews and complaints
2.2.11 Administrator management of users and services

2.3 User Classes and Characteristics

CareConnect has four main types of users. Each user has different needs and uses different features of the system.

User Class	Technical Comfort	Key Characteristics
Elderly User	Low	Can request help, check service status, send SOS alerts and contact family.
Family Member	Medium to High	Can monitor elderly users, create service requests and receive updates.
Service Provider	Medium	Can manage their profile, accept service requests and update service status.
System Administrator	High	Can manage users, verify service providers, manage services and handle complaints.

The Elderly User and Family Member are the main users of CareConnect, so the system should be simple and easy for them to use.

2.4 Operating Environment

CareConnect will be developed as a web application.

2.4.1 **Frontend:** HTML, CSS, Bootstrap, and JavaScript
2.4.2 **Backend:** Laravel and PHP

2.4.3 **Database:** MySQL
2.4.4 **Development Environment:** Laragon
2.4.5 **Browsers:** Chrome, Firefox, Edge, and Safari
2.4.6 **Devices:** Computer, laptop, smartphone, and tablet
2.4.7 **Internet:** Required to use the system

2.5 Design and Implementation Constraints

The system will use the required technologies for the project.

2.5.1 HTML, CSS, Bootstrap, and JavaScript for the frontend
2.5.2 Laravel and PHP for the backend
2.5.3 MySQL for the database

The payment system will be for tracking purposes only and a real payment gateway is not included in the prototype. Service Providers must be approved by the Administrator before providing services. The SOS feature will notify family members or emergency contacts but will not replace ambulance or medical services.

2.6 User Documentation

Simple user guides will be provided for:

2.6.1 Elderly Users
2.6.2 Family Members
2.6.3 Service Providers
2.6.4 Administrators

The guides will explain how to use the main features of the system.

2.7 Assumptions and Dependencies

The system assumes that users have an internet connection and a device with a modern web browser. Elderly Users may receive help from family members during account setup. Some features, such as location sharing, notifications, and video calling, may depend on external services or technologies.

3. External Interface Requirements

3.1 User Interfaces

CareConnect will have a simple and clean website design. We will use Bootstrap 5 to make the pages easy to use and keep the same design throughout the website. Every page will have the CareConnect logo, logout option, a simple menu and a support option. The elderly user's pages will be designed to be very simple. The buttons will be large, the text will be short and easy to understand and the colors will be clear. Common tasks such as requesting a service should take

only a few steps. The website will work on computers, smartphones and tablets. Users can use both a touchscreen and a keyboard. The main pages will be:

3.1.1 Homepage
3.1.2 Registration
3.1.3 Login
3.1.4 Elderly User Dashboard
3.1.5 Family Member Dashboard
3.1.6 Service Provider Dashboard
3.1.7 Admin Dashboard
3.1.8 Service Request
3.1.9 Provider Matching
3.1.10 Service Tracking
3.1.11 SOS
3.1.12 Video Call
3.1.13 Payment
3.1.14 Rating and Review
3.1.15 Complaint

3.2 Hardware Interfaces

CareConnect does not need any special hardware. It can work with normal devices.

3.2.1 Desktop/Laptop: Mainly used by Family Members and Admins.
3.2.2 Smartphone/Tablet: Can be used by all users.
3.2.3 Web Server: Runs the CareConnect website.
3.2.4 Database Server: Stores the system information.
3.2.5 Camera and Microphone: Needed for video calls.
3.2.6 GPS/Location: Used to find nearby service providers and share location during SOS.
3.2.7 Internet: Wi-Fi or mobile data is required.

If the camera, microphone or location is not available, the system will show a simple message to the user.

3.3 Software Interfaces

CareConnect will use these software and tools:

Software	**Purpose**
Laravel 10.x / PHP 8.x	Used for the backend and main system logic
MySQL 8.0	Used to store user and system data
Bootstrap 5	Used to design the website

Google Maps API (Optional)	Used for location and finding nearby providers
Firebase Cloud Messaging (Optional)	Used for sending notifications
WebRTC (Optional)	Used for video calls
Git / GitHub	Used to store and manage project code

3.4 Communications Interfaces

CareConnect will use normal and secure communication methods.

3.4.1 The website will use HTTPS to protect user information.
3.4.2 If frontend and backend are separate, they will communicate through REST API using JSON.
3.4.3 Notifications may be sent through a notification service.
3.4.4 Video calls will use WebRTC or a similar technology.
3.4.5 The system will show the connection status during a video call.
3.4.6 Email service is not required for the first version.

4. System Features

CareConnect has nine main features. These features explain what each type of user can do in the system.

4.1 Account Registration and Access Control

4.1.1 Description and Priority

This feature manages the service cost and payment status. After completing a service, users can give a rating and write a review. Users can also submit a complaint which the Admin can check and solve. Priority: Medium to High.

4.1.2 Stimulus/Response Sequences

When a user starts registration, they first select their role and the system shows the information they need to provide. After the user submits the form, the system checks the information and creates the account if everything is correct. When the user logs in, the system checks their email or phone number and password and then opens the correct dashboard based on their role. If the user forgets their password, the system provides an option to reset it.

4.1.3 Functional Requirements

REQ-1.1: The system shall allow users to register as an Elderly User, Family Member or Service Provider.

REQ-1.2: The system shall not allow a user to register with an email or phone number that is already used.

REQ-1.3: The system shall store passwords securely and not as plain text.

REQ-1.4: The system shall allow users to change their password after entering their current password.

REQ-1.5: The system shall allow users to reset a forgotten password using a verified email or phone number.

REQ-1.6: The system shall allow users to access only the features available for their role.

REQ-1.7: The system shall temporarily block an account after too many failed login attempts.

4.2 Profile Management and Family-Elder Linking

4.2.1 Description and Priority

This feature allows users to manage their profile information. It also allows a Family Member to connect with an Elderly User after getting permission from the Elderly User.

Priority: High

4.2.2 Stimulus/Response Sequences

When a user opens their profile, the system shows their current information. If the user changes any information, the system checks it and saves the updated information. A Family Member can also send a connection request to an Elderly User. The Elderly User can accept or reject the request, and the system will update the connection based on their choice.

4.2.3 Functional Requirements

REQ-2.1: The system shall allow an Elderly User to update their name, phone number, address, and other profile information.

REQ-2.2: The system shall allow a Family Member to update their own profile and notification settings.

REQ-2.3: The system shall allow a Family Member to send a connection request using an invitation code.

REQ-2.4: The system shall not allow a Family Member to see an Elderly User's information before approval.

REQ-2.5: The system shall allow an Elderly User to remove a Family Member from their account at any time.

4.3 Service Provider Verification and Categories

4.3.1 Description and Priority

This feature allows Service Providers to register and select the services they want to provide. An Admin will check their information before approving them. Priority: High

4.3.2 Stimulus/Response Sequences

When a Service Provider creates an account and chooses the services they want to provide, the account is set to Pending. The provider then gives their information and qualification documents. The Admin checks the information and approves or rejects the provider. After approval, the provider can add their available days and times and the system saves the schedule.

4.3.3 Functional Requirements

REQ-3.1: The system shall allow Service Providers to register and select one or more service categories.

REQ-3.2: The system shall require Admin approval before a provider can accept a service request.

REQ-3.3: The system shall require additional documents for sensitive care services.

REQ-3.4: The system shall allow the Admin to add, edit, or remove service categories.

REQ-3.5: The system shall allow providers to set and update their available time.

REQ-3.6: The system shall allow the Admin to suspend or reject a provider when necessary.

4.4 Service Request Creation and Matching

4.4.1 Description and Priority

This feature allows an Elderly User or an approved Family Member to request help. The system will find suitable providers based on their service type, available time and location. Priority: High

4.4.2 Stimulus/Response Sequences

The user fills out the service request form, and the system checks the information and creates the request. Then the system looks for suitable service providers and shows the available providers to the user. The provider checks the request and decides whether to accept or reject it. If the provider accepts the request, the system assigns the provider to that service request.

4.4.3 Functional Requirements

REQ-4.1: The system shall allow an Elderly User or authorized Family Member to create a service request.

REQ-4.2: The system shall reject a request if required information is missing or the date/time is invalid.

REQ-4.3: The system shall only show providers who are approved for the requested service.

REQ-4.4: The system shall not automatically assign a provider. The provider must accept the request.

REQ-4.5: The system shall search for another provider if a provider rejects or does not respond.

REQ-4.6: The system shall inform the user if no suitable provider is available.

4.5 Service Tracking and Family Monitoring

4.5.1 Description and Priority

This feature allows providers to update the progress of a service. Elderly Users and approved Family Members can see the current service status. Priority: High

4.5.2 Stimulus/Response Sequences

The provider updates the service status and the system saves the status with the time. The Elderly User can check the service tracking to see the current status and provider information. The linked Family Member can also check the service and see the information about the Elderly User's service.

4.5.3 Functional Requirements

REQ-5.1: The system shall allow the assigned provider to update the service status.

REQ-5.2: The system shall save every status change with date and time.

REQ-5.3: The system shall show the current status and provider information to the Elderly User and approved Family Member.

REQ-5.4: The system shall only allow Family Members to see information about Elderly Users they are connected to.

REQ-5.5: The system shall keep completed, cancelled, and disputed services in the service history.

4.6 SOS Emergency Alert and Emergency Contacts

4.6.1 Description and Priority

This feature allows an Elderly User to quickly send an SOS alert to their Family Members and emergency contacts. The user's location can also be included if location permission is available. The SOS feature is only an alert system. It is not an ambulance or medical service. Priority: High

4.6.2 Stimulus/Response Sequences

When the Elderly User presses the SOS button, the system asks for confirmation. If the user confirms, the system sends the SOS alert to their family and emergency contacts. If the location is available, the system also sends the user's location. The user can add, change, or remove emergency contacts, and the system saves the changes.

4.6.3 Functional Requirements

REQ-6.1: The system shall provide a clearly visible SOS button for Elderly Users.

REQ-6.2: The system shall ask for confirmation before sending an SOS alert.

REQ-6.3: The system shall notify linked Family Members and emergency contacts.

REQ-6.4: The system shall include the user's location only when location permission is available.

REQ-6.5: The system shall clearly state that SOS is not a replacement for an ambulance or emergency service.

REQ-6.6: The system shall allow users to add, edit, and remove emergency contacts.

4.7 Video Calling

4.7.1 Description and Priority

This feature allows an Elderly User to easily make a video call with a linked Family Member. Family Members can also call the Elderly User. Priority: Medium to High

4.7.2 Stimulus/Response Sequences

When the Elderly User presses the “Call My Family” button, the system sends a call request to the Family Member. If the Family Member accepts the call, the video call starts. A Family Member can also call the Elderly User, and the Elderly User can accept or reject the call. If needed, a caregiver can help the Elderly User make a video call during a service visit.

4.7.3 Functional Requirements

REQ-7.1: The system shall provide a simple button for an Elderly User to start a video call.

REQ-7.2: The system shall allow Family Members to call linked Elderly Users.

REQ-7.3: The system shall not automatically record video calls.

REQ-7.4: The caregiver's camera shall not stay active when there is no active call.

REQ-7.5: The system shall show a missed-call notification if a call is not answered.

REQ-7.6: The system should support audio-only calls when video does not work properly, if possible.

4.8 Payments, Ratings, and Complaints

4.8.1 Description and Priority

This feature manages service payments, ratings, reviews, and complaints. Priority: Medium to High

4.8.2 Stimulus/Response Sequences

The system shows the service cost and updates the payment status. After the service is completed, the user can give a rating and write a review. If the user has a problem, they can submit a complaint, and the system sends it to the Admin. The Admin checks the complaint and updates its status.

4.8.3 Functional Requirements

REQ-8.1: The system shall show the service cost and payment status.

REQ-8.2: The system shall keep the payment history for each service.

REQ-8.3: The system shall allow users to rate and review a service only after it is completed.

REQ-8.4: The system shall update the provider's average rating when a new review is added.

REQ-8.5: The system shall allow users to submit a complaint about a service or provider.

REQ-8.6: The system shall allow the Admin to review and resolve complaints.

REQ-8.7: The system shall allow providers to see their earnings from completed services.

4.9 Administration and Reporting

4.9.1 Description and Priority

This feature allows the Admin to manage users, providers, services, requests, and complaints. The Admin can also create reports. Priority: Medium to High

4.9.2 Stimulus/Response Sequences

The Admin can search for a user or service provider, and the system shows their account information. If the Admin suspends or deactivates an account, the system updates the account status. The Admin can also select a report, and the system shows the requested report.

4.9.3 Functional Requirements

REQ-9.1: The system shall allow the Admin to search, view, suspend, or deactivate Elderly User and Family Member accounts.

REQ-9.2: The system shall allow the Admin to search, verify, suspend, or deactivate Service Provider accounts.

REQ-9.3: The system shall allow the Admin to review service requests and handle disputed requests.

REQ-9.4: The system shall record important Admin actions in an audit log.

REQ-9.5: The system shall allow the Admin to create reports about service requests, providers, complaints, and payments.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

CareConnect should be fast enough for normal use.

5.1.1 Provider matching should normally show results within 3 seconds.
5.1.2 Dashboard pages should normally load within 2 seconds.
5.1.3 The prototype should support at least 100 users and 500 service requests for testing.

5.2 Safety Requirements

5.2.1 The system shall not provide medical treatment or medical advice.
5.2.2 Providers cannot accept sensitive care services without Admin approval.
5.2.3 The SOS message shall clearly explain that it only alerts family and emergency contacts.
5.2.4 The system shall warn the Elderly User if they have not added an emergency contact.

5.3 Security Requirements

5.3.1 Passwords shall be stored securely using password hashing.
5.3.2 All website communication shall use HTTPS.
5.3.3 Users must log in before accessing their dashboard.
5.3.4 Users can only access features allowed for their role.
5.3.5 Family Members cannot see Elderly User information without approval.
5.3.6 The system shall protect against common attacks such as SQL injection and XSS.
5.3.7 Important Admin actions shall be recorded.

5.4 Software Quality Attributes

Quality	Meaning for CareConnect
Usability	The website should be simple and easy to use, especially for Elderly Users.
Reliability	Service requests and updates should be saved correctly.
Maintainability	The code should be organized and easy to understand and change.

Portability	The system should work on common browsers and normal servers.
Availability	The system should work properly during testing and demonstrations.
Interoperability	The system should be able to work with services such as maps, notifications, and video calls.

5.5 Business Rules

5.5.1 Only the Admin can approve, reject, or suspend a Service Provider.

5.5.2 A Family Member can only access information about an Elderly User who has approved the connection.

5.5.3 A Service Provider can only accept requests for approved service categories.
5.5.4 Sensitive care services require extra qualification documents.
5.5.5 A service can only be rated once and only after completion.
5.5.6 An Elderly User can remove a Family Member's access at any time.

6. Other Requirements

6.1 Database Requirements

CareConnect will use MySQL to store all important system information.

The database will store:

6.1.1 User information
6.1.2 Profiles
6.1.3 Service categories
6.1.4 Service requests
6.1.5 Service providers
6.1.6 Payments
6.1.7 Reviews
6.1.8 Complaints
6.1.9 Notifications
6.1.10 Emergency contacts
6.1.11 SOS alerts
6.1.12 Video call information
6.1.13 Admin actions

6.2 Legal and Compliance Notes

6.2.1 CareConnect will clearly state that it is a service coordination platform not a medical or ambulance service.

6.2.2 User information such as name, phone number and location will only be used for the purposes of the system, such as service matching, tracking and safety alerts.

6.3 Internationalization

The first version of CareConnect will be developed in English. The system should be designed so that other languages can be added in the future.

6.4 Reuse Objectives

The project will use existing and well-known tools instead of creating everything from the beginning. For example:

6.4.1 Laravel
6.4.2 Bootstrap
6.4.3 Laravel Authentication
6.4.4 PHP packages
6.4.5 MySQL

This will save development time and make the project easier to maintain.

Appendix A: Glossary

Term	Meaning
Elderly User	A person who uses CareConnect to request help.
Family Member	A person connected to an Elderly User and allowed to see their information.
Service Provider	A person who provides services through CareConnect.
Administrator	A person who manages the CareConnect system and checks providers.
Service Category	A type of service, such as General Helper or Caregiver.
Verification	The process of checking a Service Provider before approval.
SOS Alert	An emergency alert sent to family members and emergency contacts.
REQ	A number used to identify a specific requirement.
RBAC	Role-Based Access Control. It means users can access features based on their role.
API	A way for two software systems to communicate with each other.

ERD	A diagram that shows the relationships between database tables.
TBD	To Be Determined. It means something has not been decided yet.

Appendix B: Analysis Models

The following diagrams will be created separately using tools such as Draw.io, Figma, Lucidchart or Visual Paradigm:

1. **ERD:** Shows the main database tables and their relationships.
2. **Use Case Diagram:** Shows the four types of users and what they can do.
3. **Class Diagram:** Shows the main classes and their relationships.
4. **Activity Diagram:** Shows the steps of creating and completing a service request.
5. **Sequence Diagram:** Shows how the system works when a request is created, matched, accepted, and completed.
6. **State Diagram:** Shows the different stages of a service request, such as Pending, Searching, Accepted, Assigned, On the Way, Started, In Progress, Completed, Cancelled, and Disputed.

Appendix C: To Be Determined List

TBD	**Open Item**
TBD1	Which payment gateway will be used, if any, in the prototype.
TBD2	Which video calling API or library will be used.
TBD3	Whether Firebase notifications will be fully implemented or simulated during the demonstration.
TBD4	The final terms of use and privacy notice for the system.
TBD5	What documents will be required from a provider applying for the Professional/Experienced Elder Caregiver category.